Theorem C.2 as a problem

Problem. *Use the notation of Appendix C. Thus $f_{0,m}$ denotes the balanced Hankel determinant, $f_{1,m}$ denotes the first finite-size correction polynomial, and g_m denotes the non-zero threshold factor defined by*

$$d_{m+1}(\lambda) = \lambda^{\lceil (m+1)/2 \rceil} g_m(\lambda), \qquad g_m(0) \neq 0.$$

Theorem C.2. *For every $m \geq 0$,*

$$g_m(\lambda) \mid \bigl(f_{1,m}(\lambda) - f'_{0,m}(\lambda)\bigr) \qquad \text{in } \mathbb{Q}[\lambda].$$